OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 26 OF 55

Network Intrusion Detection Using Ensemble Machine Learning and Anomaly Detection

Domain: Network Security & Cyber Anomaly Detection | **Module:** project_26_network_intrusion

OFFICIAL ABSTRACT

The system is developed using Python, machine learning algorithms, and anomaly detection techniques to analyze network traffic and identify suspicious behavior. Relevant network features such as connection patterns, traffic volume, protocol information, duration, and packet-related characteristics are processed and provided to the learning models. Ensemble machine learning techniques combine the predictions of multiple models to improve detection reliability compared with depending on a single algorithm. The major uniqueness of the proposed system is the combination of ensemble-based classification with anomaly detection. While the classification component can identify known attack patterns, the anomaly detection component focuses on recognizing network behavior that significantly differs from normal traffic. This provides an additional layer of protection against unusual or previously unseen activities that may not match existing attack signatures. The system can also assign a risk level to detected network activities, allowing security administrators to identify potentially serious events more quickly. By analyzing traffic continuously, the proposed approach can support early identification of suspicious behavior and reduce the dependency on manually defined security rules. The model can be trained and evaluated using suitable network traffic datasets containing both normal and attack-related activities. The proposed solution can be applied in organizations, educational institutions, cloud environments, enterprise networks, and other systems where continuous network monitoring is required. Overall, the project demonstrates how ensemble machine learning and anomaly detection can work together to create a more adaptive intrusion detection system, capable of identifying known attacks as well as abnormal network behavior and improving the overall effectiveness of network security monitoring.

📌 Project Summary & System Specifications

Project Title	Network Intrusion Detection Using Ensemble Machine Learning and Anomaly Detection
Specialization / Category	Network Security & Cyber Anomaly Detection
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5026`
Local Implementation	`project_26_network_intrusion`